

Problem 1 (7.21). Let

$$G(x, y) = \left(\frac{x}{x^2 + y^2}, \frac{y}{x^2 + y^2} \right), \quad F(x, y) = \left(\frac{-y}{x^2 + y^2}, \frac{x}{x^2 + y^2} \right)$$

- (a) Find a potential function $g(x, y)$ so that $\nabla g = G$. What is the domain of g ?
- (b) Is $\tan^{-1}(y/x)$ is a potential function for F ? Justify your answer.

Problem 2 (7.42). Find the flux of the vector field $F = (2y, 3z, 4x)$ through surface $D = [0, 1] \times \{0\} \times [0, 1]$.

Problem 3 (7.50). Let S be the sphere of radius $R > 0$ centered at the origin of $\mathbb{R}^3$. Consider a vector field of the form

$$F(X) = k(\|X\|)X$$

where k is a function only of the norm $\|X\|$. Suppose the outward flux $\int_S F \cdot N d\sigma$ does not depend on the radius of S . Prove that $k(\|X\|)$ is some multiple of $\|X\|^{-3}$.

Problem 4 (7.51). Suppose surface S is parametrized by $X(u, v)$, (u, v) in some smoothly bounded set D , and k is a positive number. Let T be the set of points

$$Y(u, v) = kX(u, v), \quad (u, v) \text{ in } D.$$

Show that T is a smooth surface parametrized by Y . Also, show that $\text{Area}(T) = k^2 \text{Area}(S)$.